

LITERATURE REVIEW

TEAM: FactCheck AI

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Sno.	Reference	Problem addressed	Methodology/technique	Dataset	Results/performance	Limitations	Research gap
1.	SelfCheckGPT (Manakul et al., 2023)	Detecting hallucinations in LLM outputs without external knowledge	Self-consistency check using multiple generations	Wikibio and GPT-3 generated biographies	Outperforms baselines in hallucination detection	Relies on model self-consistency and may miss subtle errors	Uses only self-consistency for hallucination detection and evaluates a limited set of LLMs
2.	HaluEval (Li et al., 2023)	Lack of a standardized hallucination evaluation benchmark	Benchmark across QA, summarization, dialogue	HaluEval dataset (multi-domain tasks)	Reveals high hallucination rates across LLMs	Limited to English; benchmark scope is narrow	Provides a benchmark dataset but does not build an automated framework using machine learning.
3.	FActScore (Min et al., 2023)	Difficulty in evaluating factuality of long-form LLM outputs	Atomic fact decomposition and retrieval-based scoring	Biographies, 6,500 LLM generations	ChatGPT approx 58% factuality; automated metric is <2% error vs human	Focused on biographies; limited domain coverage	Focuses on factuality scoring rather than comparative evaluation of different LLMs.
4	SAFE (Wei et al., 2024)	Costly human evaluation of factuality	Search-Augmented Factuality Evaluator (LLM + Google Search)	LongFact benchmark (38 topics, 16k facts)	72% agreement with humans; 20x cheaper	Relies on search engine availability; recall trade-offs	Depends on external web search for verification and does not compare several LLMs under identical conditions
5	TruthfulQA (Lin et al., 2021)	Models mimic human falsehoods	Benchmark with 817 questions across 38 categories	TruthfulQA dataset	Best model truthful on 58% vs humans 94%	benchmark is limited in scope	Introduces a benchmark but does not provide an automated evaluation pipeline for comparing multiple LLMs
6	RAGAS (Es et al., 2023)	Evaluating Retrieval-Augmented Generation pipelines	Reference-free evaluation metrics	RAG pipelines	Enables evaluation without ground truth	Metrics miss nuanced errors	Designed primarily for RAG systems and is not intended for general LLM hallucination comparison

7	Factored Verification (George & Stuhlmüller, 2023)	Hallucinations in summarizing academic papers	Automated hallucination detection as well as self-correction	HaluEval benchmark	76.2% accuracy; reduced hallucinations in GPT-4	Focused on academic summarization only	Focuses on hallucination detection in summarization tasks only. A generalized evaluation framework is required.
8	SciFact-Open (Wadden et al., 2022)	Scientific claim verification in open-domain	Pooling and annotation of evidence	500K research abstracts	Performance drop ≥ 15 F1 vs smaller corpora	Evidence annotation is incomplete	Targets scientific claim verification only. There is a need for a domain-independent framework .
9	FEVER (Thorne et al., 2018)	Large-scale claim verification	Wikipedia-based claims, annotated evidence	185,445 claims	Best accuracy 31.87% with evidence	Limited to Wikipedia; shallow claims	Provides a large fact-verification benchmark but does not evaluate generative AI models
10	FaithBench (Bao et al., 2024)	Lack of diverse hallucination benchmarks	Benchmark with 10 modern LLMs, 8 families	Summarization tasks	GPT-4o least hallucinations; detection models approx. 50% accuracy	Detection models weak; benchmark limited to summarization	does not include an end-to-end evaluation system with automatic hallucination prediction
11	FELM (Chen et al., 2023)	Evaluating factuality evaluators themselves	Benchmark with fine-grained annotations	Diverse domains	Retrieval helps but evaluators still weak	Evaluators are not reliable	Evaluates factuality assessment methods but does not compare multiple state-of-the-art LLMs
12	Survey (et al., 2024)	Hallucination mitigation overview	Taxonomy of 32+ techniques	Multiple datasets	Categorization of methods (RAG, CoNLI, CoVe, etc.)	Survey only	Need standardized evaluation across techniques

The Overall Research Gap

From the literature review, it is evident that existing studies either introduce hallucination benchmarks, evaluate individual LLMs, or propose specific factuality assessment methods. However, there is limited work on a unified, automated framework that compares multiple state-of-the-art generative AI models using a common set of fact-checking prompts and employs a lightweight machine learning model to automatically predict whether responses are factual or hallucinated. Furthermore, most existing approaches lack a deployable interface that enables practical comparison of LLM performance. This project aims to address these limitations by developing an end-to-end evaluation framework that integrates multi-model benchmarking, automated hallucination prediction, and an interactive dashboard.